

Publication List – Yuzhe Zhang

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Personal Information

Address: GSI Helmholtz Institute Mainz, Staudingerweg 18, 55128 Mainz, Germany

Email: yuhzhang@uni-mainz.de

ORCID: [0009-0006-6082-9066](https://orcid.org/0009-0006-6082-9066)

Google Scholar: scholar.google.com/citations?user=tPAwEPsAAAAJ

Journal Articles

1. Y. Zhang, D. A. Tumturk, H. Bekker, D. Budker, D. F. J. Kimball, A. O. Sushkov, and A. Wickenbrock, “Frequency-Scanning Considerations in Axionlike Dark Matter Spin-Precession Experiments,” *Annalen der Physik*, vol. 536, no. 1, p. 2 300 223, 2024
2. J. Walter, H. Bekker, J. Blanchard, D. Budker, N. L. Figueroa, A. Wickenbrock, Y. Zhang, and P. Zhou, “Fast shimming algorithm based on bayesian optimization for magnetic resonance based dark matter search,” *Annalen der Physik*, vol. 536, no. 1, p. 2 300 258, 2024. DOI: <https://doi.org/10.1002/andp.202300258>
3. J. Walter, O. Maliaka, Y. Zhang, J. W. Blanchard, G. Centers, A. Dogan, M. Engler, N. L. Figueroa, Y. Kim, D. F. J. Kimball, M. Lawson, D. W. Smith, A. O. Sushkov, D. Budker, H. Bekker, and A. Wickenbrock, “Search for axionlike dark matter using liquid-state nuclear magnetic resonance,” *Phys. Rev. D*, vol. 112, p. 052 008, 5 Sep. 2025. DOI: [10.1103/39nc-vr9m](https://doi.org/10.1103/39nc-vr9m)
4. D. Aybas, H. Bekker, D. Budker, W. Ji, O. Kim, Y. Kim, D. F. Jackson Kimball, J. Liu, X. Ma, C. P. Salemi, Y. K. Semertzidis, A. O. Sushkov, K. Wei, A. Wickenbrock, and Y. Zhang, “Cavity, lumped circuit, and spin-based detection of axion dark matter: Differences and similarities,” *Universe*, vol. 12, no. 4, 2026, ISSN: 2218-1997. DOI: [10.3390/universe12040106](https://doi.org/10.3390/universe12040106)
5. Y. Xing, X. Kong, X. Guo, Y. Liu, Q. Li, Y. Zhang, Y. Sheng, X. Yang, Z. Geng, and J. Zeng, “Bi@sn core-shell structure with compressive strain boosts the electroreduction of co(2) into formic acid,” *Advanced science (Weinheim, Baden-Wuerttemberg, Germany)*, vol. 7, no. 22, p. 1 902 989, 2020, ISSN: 2198-3844 (Print). DOI: [10.1002/advs.201902989](https://doi.org/10.1002/advs.201902989)

Preprint

1. Y. Zhang, *Axionbloch: An open-source python package for simulating axion-induced spin dynamics*, 2026

Software

1. Y. Zhang, *Axionbloch*, version 0.2.7, Open-source software repository, GitHub, 2026

Educational Publication

1. Y. Zhang and D. Budker, *Some rules of good scientific viewgraphs*, Zenodo, 2024. DOI: [10.5281/zenodo.13134573](https://doi.org/10.5281/zenodo.13134573)
2. D. Budker, D. Jackson Kimball, A. Trabesinger, and Y. Zhang, 科研报告的规矩方圆 *Some Rules for Good Scientific Presentations*. Zenodo, Feb. 2026. DOI: [10.5281/zenodo.18644045](https://doi.org/10.5281/zenodo.18644045)

Manuscripts in Preparation

1. J. Gao, Y. Zhang, J. Walter, and D. Budker, “Search for earth bound axionlike dark matter,” in preparation
2. Y. Zhang, J. Walter, O. Maliaka, and D. Budker, “What can casper-gradient do,” in preparation